Introduction To Quantum Technologies and Applications
Assignment-3

1Q.why are the requirements to build a quantum computer explain?
A.Classical Computer (Traditional)

- Uses classical bits (0 or 1)
- Built with well-established semiconductor technology
- Works reliably at room temperature
- Solves many everyday problems efficiently
- Limited for certain complex problems (e.g., large-scale simulations, optimization)

Quantum Computer (Next Generation)

- Uses quantum bits (qubits)
- Exploits superposition, entanglement and quantum interference
- Requires extreme environmental control (very low temperature, isolation, etc.)
- Has the potential to solve certain problems exponentially faster
- Still a major research and engineering challenge

To build a practical quantum computer, we need more than just quantum chips.

A working quantum computer requires a combination of physical, theoretical, and engineering advances.
The most widely accepted guidelines are given by the DiVincenzo criteria.

Key Requirements

1. Well-defined Qubits
Physical systems that can reliably represent quantum states.

2. Initialization
Ability to prepare qubits in a known initial state (usually $|0\rangle$).

3. Long Decoherence Time
Qubits must retain their quantum state long enough to perform computations.

4. Universal Quantum Gates
Ability to perform a set of gates to carry out any quantum computation.

5. Measurement
Ability to accurately measure the final quantum state and obtain classical results.

Importance

- Quantum computers can potentially solve problems that are intractable for classical computers.
- They can impact fields such as cryptography, drug discovery, materials science, optimization, and artificial intelligence.
- But achieving a large-scale, reliable quantum computer requires meeting all the above requirements simultaneously.

Real-World Applications (Potential)

- Breaking classical encryption (e.g., RSA)
- Designing new drugs and materials
- Solving complex optimization problems
- Simulating quantum systems (chemistry, physics)
- Machine learning and data analysis

A quantum computer is not just a faster computer, but a fundamentally new way of processing information.

2. DiVincenzo Criteria for a Quantum Computer

A physical system can be used to build a quantum computer if it satisfies the following five criteria, proposed by David P. DiVincenzo (2000).

1. Well-defined Qubits

Physical systems that can reliably represent quantum bits (qubits) with two distinguishable states $|0\rangle$ and $|1\rangle$.

- Must be a two-level quantum system.
- States should be distinguishable and controllable.
- Examples: superconducting circuits, trapped ions, photons, etc.

2. Initialization

Ability to prepare the qubits in a known initial state, usually the $|0\rangle$ state.

- All qubits must start from a well-defined state.
- High-fidelity initialization is required.

- Often achieved by cooling or optical pumping depending on the physical platform.

3. Long Decoherence Time

The qubits must retain their quantum state long enough to perform useful computations before losing coherence.

- Decoherence time (T_2) should be much longer than the gate operation time.
- Long coherence allows deeper circuits.
- Isolation from the environment is important.

4. Universal Quantum Gates

Ability to perform a set of quantum gates that can carry out any quantum computation (i.e., universal set).

- A small set of gates (e.g., H, CNOT) is sufficient for universality.
- Must be able to perform single-qubit and multi-qubit gates.
- Gates should be high-fidelity and precisely controllable.

5. Measurement

Ability to accurately measure the final quantum state and obtain classical results.

- Must be able to distinguish between $|0\rangle$ and $|1\rangle$ with high accuracy.
- Measurement collapses the quantum state to a classical bit.
- High readout fidelity is essential for reliable computation.

Q2.Why maintaining Entanglement is difficult?

A.Entanglement is one of the most important and unique features of quantum mechanics. It provides strong correlations between two or more qubits and is a key resource for quantum computing.

1. What is Quantum Entanglement?

- When two or more qubits are entangled, their quantum states are correlated, no matter how far apart they are.
- The measurement result of one qubit is instantly related to the result of the other qubit(s).

- Entanglement is a purely quantum phenomenon and has no classical counterpart.

Entangled qubits share a single quantum state and cannot be described independently.

2. Example: Bell State (Two-Qubit Entanglement)

A common example of an entangled state is the Bell state:

$|\Phi^+\rangle = 1/\sqrt{2}\ (|00\rangle + |11\rangle)$

- If we measure the first qubit and get 0, the second qubit will be 0.
- If we measure the first qubit and get 1, the second qubit will be 1.
- The results are perfectly correlated, even if the qubits are far apart.
- Neither qubit has a definite state (0 or 1) before measurement; the pair is in a superposition.

3. Illustration of Entangled Qubits

Qubit A
State of A is not definite until measurement.

Qubit B
State of B is not definite until measurement.

Measurement of one qubit instantly determines the state of the other qubit, regardless of the distance between them.

4. Measurement Correlations

Measurement on Qubit A	Result on Qubit B	Correlation
0	0	Perfectly correlated
1	1	Perfectly correlated

Entanglement enables non-classical correlations that are essential for many quantum algorithms and protocols.

5. Effect of Decoherence on Entanglement

Entanglement is very sensitive to environmental interactions.
Decoherence can destroy the entangled state and reduce the quantum correlations.

Maintaining entanglement for a sufficient time is a major challenge in building practical quantum computers.

6. Importance in Quantum Computing

- Entanglement is a key resource for quantum speedup.
- Used in many quantum algorithms (e.g., Shor's algorithm, Grover's algorithm, quantum simulation).
- Essential for quantum error correction (logical qubits).
- Enables quantum communication protocols (e.g., quantum teleportation, superdense coding).
- More complex entanglement (multi-qubit entanglement) is needed for large-scale quantum computation.

Q3.Discuss fragility of Quantum system?

A.Quantum systems are very fragile. Even small changes in the surroundings can make them lose their quantum properties. The main reasons are decoherence, noise and the difficulty of control.

1. A Qubit in Its Ideal State

An isolated qubit can be in a superposition of $|0\rangle$ and $|1\rangle$.

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

The qubit maintains a well-defined quantum state with a fixed amplitude and phase.

2. Interaction with Environment

In reality, a qubit cannot be completely isolated. It interacts with its environment, leading to loss of quantum coherence.

- Temperature fluctuations (heat)
- Electromagnetic radiation (EM waves)
- Vibrations (mechanical disturbances)
- Material imperfections
- Other qubits (crosstalk)

3. Decoherence

When the qubit interacts with the environment, its superposition state gradually loses coherence and behaves like a classical bit.

Initial state (superposition) → Loss of coherence (reduction of quantum properties) → Final state (classical 0 or 1)

4. Types of Noise

Noise refers to unwanted disturbances that cause errors in quantum information.

- Thermal noise – due to temperature changes
- Electromagnetic noise – from external fields, electronics, etc.
- Control noise – imperfections in control pulses
- Photon/radiation noise – from background radiation
- Crosstalk – interaction with nearby qubits
- Material noise – due to defects in the hardware

5. Effect of Noise on Qubit State

Noise can change the amplitude and phase of the qubit, leading to errors.

The qubit state becomes uncertain due to noise.

6. Control Challenge

We need to protect the qubits from the environment, but at the same time we must control them using external signals (lasers, microwave pulses, etc.). This is a major challenge.

Isolation (protect from environment)

Control (apply precise operations)

Quantum engineering aims to achieve a balance between isolation and control.

Q4.Explain scalability challenge in quantum hardware?

A.Scalability Challenge in Quantum Hardware

1. Scalability

The system should allow the addition of a large number of qubits (e.g., thousands or millions) without major difficulties.

Requirements for scalability:

- Qubit fabrication methods that can be scaled.
- Efficient control and readout architecture.
- Manage crosstalk between qubits.
- Modular design (small units connected together).
- Maintain high fidelity even as the number of qubits increases.

Scalability is necessary to solve real-world, useful problems.

2. Superconducting Quantum Computer – Challenges

- Short coherence time (typically 10 – 100 microseconds).
- Sensitive to noise and environmental disturbances.
- Requires complex cryogenic systems.
- Qubit variability and fabrication defects.
- Scaling to error-corrected, fault-tolerant systems is still an open challenge.

3. Trapped Ion Quantum Computer – Challenges

- Slower gate operations (microseconds to milliseconds).
- Difficult to scale to very large number of qubits (1000+).
- Need complex laser and vacuum systems required.
- Need precise alignment and stability.
- Sensitive to technical noise (e.g., laser fluctuations).

4. Photonic Quantum Computer – Challenges

- Photons do not interact easily with each other.
- Two-qubit gates are difficult (requiring nonlinear optics or measurement-based schemes).
- Probabilistic operations (not deterministic in many cases).
- Requires high-efficiency single-photon sources and detectors.
- Scaling to large numbers of qubits is challenging.

5. Topological Quantum Computer – Challenges

- Still in early experimental stages.
- Unambiguous detection of Majorana zero modes is difficult.
- Requires advanced materials and precise nanofabrication.
- Braiding operations are technically challenging.
- Scaling to large numbers of qubits is not yet demonstrated.

Q5.Discuss vision vs reality of quantum technologies?

A.Vision

- Quantum computers use qubits and exploit superposition, entanglement and quantum interference.
- They have the potential to solve certain problems exponentially faster.
- They can impact fields such as cryptography, drug discovery, materials science, optimization, and artificial intelligence.
- Quantum computers can potentially solve problems that are intractable for classical computers.
- Quantum error correction and scalable systems could lead to fault-tolerant, large-scale quantum computation.
- Quantum technologies can also enable quantum communication, networking and certain quantum algorithms.

Reality

- Building a practical quantum computer is still a major research and engineering challenge.
- Qubits are highly sensitive to their environment.
- Decoherence and noise can cause loss of quantum behavior and errors in quantum information.
- Precise control is required while maintaining isolation.
- Maintaining entanglement for a sufficient time is a major challenge.
- Quantum error correction is necessary because quantum states are fragile and can suffer from noise and decoherence.
- Adding a large number of qubits without major difficulties is a scalability challenge.
- Different hardware platforms have their own challenges:
    - Superconducting qubits: short coherence time, environmental noise, complex cryogenic systems and scaling to error-corrected systems.
    - Trapped ions: slower gate operations and difficulty scaling to very large numbers of qubits.
    - Photonic systems: difficulty implementing two-qubit gates and scaling to large numbers of qubits.

    - Topological systems: still in early experimental stages and scaling to large numbers of qubits is not yet demonstrated.

What is Working and What Remains Elusive

Working / Developing:

- Superconducting, trapped-ion and photonic quantum computers are established research platforms.
- Superconducting qubits can be integrated on a single chip and have fast gate operations.
- Trapped ions provide very high qubit fidelity and long coherence times.
- Photonic systems operate at room temperature and are well-suited for quantum communication.
- Quantum error correction methods and advanced hardware platforms are actively being developed.

Remains Elusive:

- A large-scale, reliable and fault-tolerant quantum computer.
- Maintaining quantum states and entanglement for sufficiently long times.
- Scaling to thousands or millions of qubits while maintaining high fidelity.
- Completely error-free quantum computation is not possible in practice, so effective error management is required.
- Topological quantum computing remains an experimental platform.

Q6.Discuss the theoretical challenges and requirements to build a quantum computer?

A.Theoretical Challenges and Requirements to Build a Quantum Computer

1. Needed to Build a Quantum Computer

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Key Requirements

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   Ability to accurately measure the final quantum state and obtain classical results.

2. Fragility of Quantum Systems

Quantum systems are very fragile. Even small changes in the surroundings can make them lose their quantum properties. The main reasons are decoherence, noise and the difficulty of control.

- Qubits are highly sensitive to their environment.
- Decoherence and noise are the main reasons for loss of quantum behavior.
- Precise control is required while maintaining isolation.
- Reducing noise and improving coherence time are critical for building practical quantum computers.

3. Conditions for a Functional Quantum System

A quantum computer will work properly only if certain conditions are satisfied. The four key conditions are isolation, error management, scalability and stability.

Isolation

Qubits must be kept completely separate from the outside world.

Error Management

Quantum systems are prone to errors due to noise and decoherence. These errors must be identified and corrected or mitigated.

Scalability

The system should allow the addition of a large number of qubits (e.g., thousands or millions) without major difficulties.

Stability

Qubits must maintain their quantum states for a sufficiently long time during computation.

4. Maintaining Quantum Entanglement

Entanglement is one of the most important and unique features of quantum mechanics. It provides strong correlations between two or more qubits and is a key resource for quantum computing.

Entanglement is very sensitive to environmental interactions. Decoherence can destroy the entangled state and reduce the quantum correlations.

Maintaining entanglement for a sufficient time is a major challenge in building practical quantum computers.

5. Quantum Error Correction

Quantum states are fragile and can suffer from errors due to noise and decoherence.

- Qubits interact with the environment, leading to bit flips, phase flips or general errors.
- Even small error rates can make long computations impossible.
- QEC allows detection and correction of errors without directly measuring the quantum information.
- The no-cloning theorem prevents simple copying of quantum states.
- Quantum error correction uses multiple entangled qubits to protect information.